Ramlakhan Madheshiya

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SUMMARY

Software engineer and B.Tech CSE graduate (2026). Active open-source contributor to the Julia scientific-computing ecosystem (SciML / OrdinaryDiffEq.jl), where I fix delay differential equation solver bugs. I also ship Godot games and offline-first web apps end to end, and have industry experience in .NET, Python and full-stack JavaScript, including an onsite internship at NTRO.

OPEN-SOURCE CONTRIBUTIONS

SciML / OrdinaryDiffEq.jl

Oct 2026 – present

Julia · high-performance ODE, DDE and DAE solvers · 2 merged, 5 open pull requests, 5 issues reported

- **Merged** #4849: allow zero constant lags and reject wrong-sign lags in DDE initialization (Unitful-safe, respects reversed `tspan`).
- **Merged** #4834: restored `default_linear_interpolation` method order so solver opt-outs apply again, keeping DAE solvers on dense output.
- **Open** #4874, #4879, #4892, #4899: fixes to DelayDiffEq's `MethodOfSteps`: fixed-point solver for `constrained` mode, `perform_step!` for SDE caches, AutoSwitch/`DefaultODEAlgorithm` handling, and stiff sub-cache types in the history integrator.
- **Open** #4872: build the discontinuity bracket in the unitless step-fraction type, enabling Unitful and Measurements time with discontinuity detection.
- Reported and root-caused DDE and nonlinear-solver issues: #4876, #4875, #4891, #4895, NonlinearSolve.jl #1372.

ibis-project / ibis

Sep 2024

Python · portable dataframe library

- #10090 (merged): made navigation-bar dropdowns scrollable in the docs.

EXPERIENCE

SDE Intern · National Technical Research Organisation (NTRO)

Apr 2025 – Nov 2025

Onsite · 3 months, in phases

- Built a security tool to improve internal compliance and operational processes (details confidential under NDA).

Web Developer Intern · Krtya Technologies Pvt. Ltd

Jul 2024 – Dec 2024

React, Node.js, Express, PostgreSQL

- Developed a scalable full-stack checkout portal with secure payments and order management.

Technical Content Writer · GeeksforGeeks

Jan 2024 – Jan 2025

Remote

- Authored 12+ technical articles, from Python and web basics (jQuery, NPM) to PyTorch and Selenium testing. Contributions page.

PROJECTS

Brainnova Studio: Godot games

Games · GitHub

GDScript, Godot 4, Android, Web, GitHub Actions

A studio site and seven mobile and browser puzzle and arcade games: Single Shot (360 solver-checked rooms, in Google Play closed testing), Ringburst, Word Glow, Few Turns, Snug Fit, Mirrorlock and Corral. Levels are generated and validated by in-repo solvers and tests; CI builds the web and Android exports and publishes them.

Bill Studio: offline billing for any country

Live · GitHub

JavaScript, PWA, IndexedDB, Web Crypto

Installable, fully offline PWA with GST, VAT and sales-tax rules for 50+ countries, a drag-and-drop bill designer, ledgers, reports (GSTR-1, P&L), payment QR codes and AES-GCM encrypted backups. No server; data stays on the device.

Group Policy Object Manager for Windows GitHub

C#, .NET, WPF · Smart India Hackathon 2024 finalist

Windows application that generates and manages customised GPOs aligned with CIS security benchmarks, built for NTRO's problem statement.

Windows CVE Analysis: Vulnerabilities, Exploits, and Defenses Paper

Manuscript · 2025 · with Santosh Saha

First-author manuscript analysing Windows CVEs, exploits and defenses.

ACHIEVEMENTS

- **Smart India Hackathon 2024 finalist**: reached the final round of India's largest national hackathon with a security tool for NTRO.

EDUCATION

Bachelor of Technology, Computer Science and Engineering 2022 – 2026

Uka Tarsadia University (UTU), Bardoli, Gujarat, India · CGPA 8.97 / 10

SKILLS

- **Languages:** Julia, Python, C, C++, C#, JavaScript, TypeScript, GDScript, HTML/CSS, Shell
- **Frameworks & tools:** Godot, Astro, .NET/WPF, React, Node.js, Flutter, Docker, Git, GitHub Actions
- **Interests:** scientific computing, numerical solvers, game design, security